CMSE 492

2024–2025 Premier League Predictions

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Background and Motivation

The Premier League is one of the most popular and competitive leagues in the world, containing a rich history

I've played competitive soccer my entire life and would consider myself an avid soccer fan

Implementing machine learning to accurately predict match outcomes and league standings

Identifying features that have the greatest impact when making predictions

Methodology

Data

League match data that includes all matches from 1993-present

Preprocessing

- Removed unnecessary columns
- Filtered matches from 2014- Present
- Handled missing values and implemented feature engineering

Features Created

- **Goal Difference**
- **Recent Form:** Average points over the last 5 matches
- **Attack & Defense Metrics:** Goals scored and conceded for both home and away teams
- **Home Advantage**
- **Time Based Features (Month and Year)**

Models

- **Baseline:** Random Forest Regression
- **Tested**:
  - Gradient Boost
  - XGBoost
  - Lasso

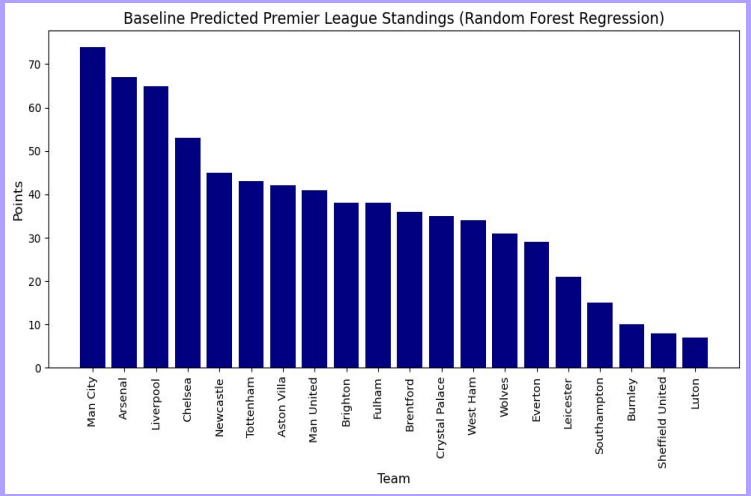
Baseline Results

Evaluation Metrics

- Mean Absolute Error: 1.064865
- Mean Squared Error: 1.786743
- Root Mean Squared Error: 1.336691
- R^2 Score: 0.515230

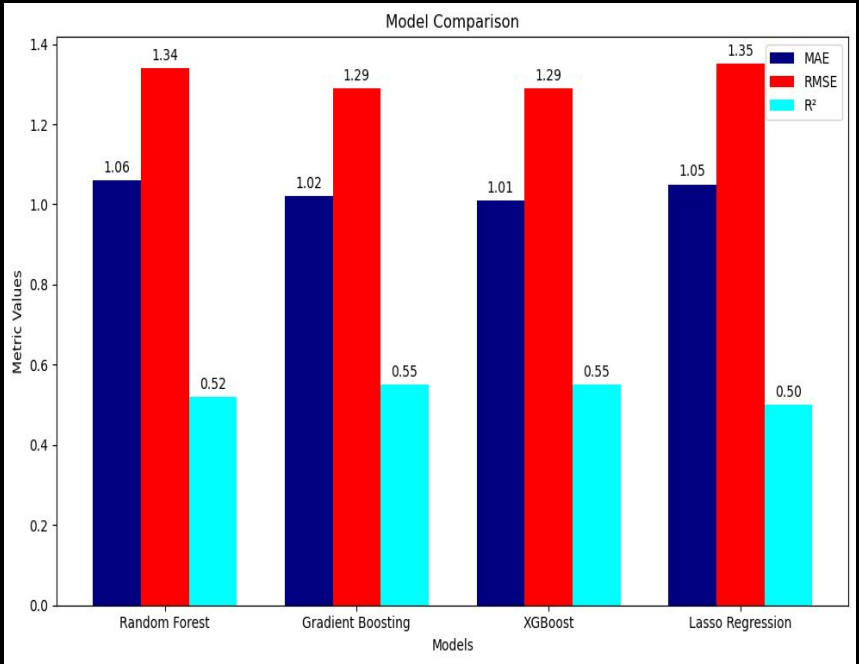
The baseline model is functional with room for improvement

The model can place well known/strong teams such as Man City (reigning champs)



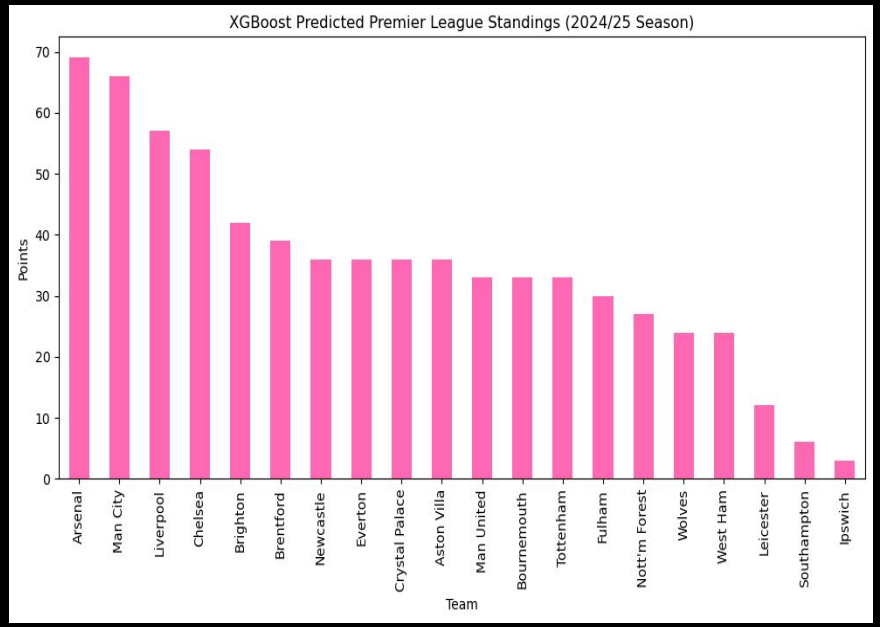
Model Comparison

- XGBoost performs the best as it has the lowest errors and highest R^2 value
- Lasso is least effective with high errors and lowest R^2
- XGBoost and Gradient Boosting are the strongest models for capturing the relationship between features and goal differences, which have a direct impact match outcomes and, therefore, season standings



Final Prediction Based on XGBoost Model

1. Arsenal	11. Man United
2. Man City	12. Bournemouth
3. Liverpool	13. Tottenham
4. Chelsea	14. Fulham
5. Brighton	15. Nottingham Forest
6. Brentford	16. Wolverhampton
7. Newcastle	17. West Ham
8. Everton	18. Leicester
9. Crystal Palace	19. Southampton
10. Aston Villa	20. Ipswich



Qualifies for Champions League

Discussion

Key Factors: The model effectively highlights key factors influencing match outcomes, such as offensive/defensive strength and home advantage

Top and Bottom Alignment: The model approximates team performance well, as the top and bottom of the standings closely align with actual league expectations.

Mid-Tier Challenges: The model struggles with mid-tier teams, often underestimating the performance of some clubs

Future Directions

- Add additional features like player performance/star players, injuries, transfer activity, and more
- Focus on fine-tuning the model to better capture the dynamics of mid-tier teams
- Try and combine multiple models to strengthen results

Thank you!